SOLUTION

SECTION–I

1. Ans. (B)

Sol. Ans. (B)

Clearly $9-x\geq 0,\ x-1\geq 0$ & $9-x-(x-1)\geq 0$

$\Rightarrow\ 1\leq x\leq 5,\ x\in I$

$\therefore\ m=5$

Range of $f(x)$ $\Rightarrow$ Domain $=\{1,2,3,4,5\}$

$\{1, 7, 15, 10\}$

$\therefore\ n=4$

2. Ans. (A)

Sol. Let weight of each ball is w gms

number of possible groups of 3 balls $= {}^nC_3$

$\Rightarrow$ total weight of ${}^nC_3$ groups $= {}^nC_3 \cdot 3w$ gms.

Similarly total weight of possible groups of 4 balls $= {}^nC_4\ 4w$ gms.

$$\Rightarrow \frac{{}^nC_4.4w}{{}^nC_3 3w}=\frac{640}{480}$$

$$\frac{n-3}{4}\cdot\frac{4}{3}=\frac{4}{3}$$

$n=7$

3. Ans. (B)

Sol. *Case-I :*

a, b, c having exactly one 5 with :

(i) 2 even digits (different or same) :

$${}^4C_2.3!+{}^4C_1.\frac{3!}{2!}=48$$

(ii) one digit divisible by 4 and one odd:

${}^2C_1.{}^4C_1.3!=48$

Case-II :

a, b, c having exactly two 5 with :

(i) one digit divisible by 4 : $2\cdot\frac{3!}{2!}=6$

Hence number of ways = 102

4. Ans. C

Sol. $(e^x-1)^2=e^{2x}-2e^x+1$

$$e^x=1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}$$

$$e^{2x}=1+\frac{2x}{1!}+\frac{(2x)^2}{2!}+\frac{(2x)^3}{3!}+\frac{(2x)^4}{4!}$$

co-efficient of $x^4=\frac{2^4}{4!}-\frac{2}{4!}=\frac{7}{12}$

5. Ans. B

Sol. k^{th} term $\rightarrow T_k=(k^2+1)\,k!$

$(k^2+1)=k^2+k-(k-1)=k(k+1)-(k-1)$

$T_k=(k^2+1)\,k!=k(k+1)!-(k-1)k!$

$T_1=1(1+1)!-0\times 1!$

$T_2=2(2+1)!-1\times 2!$

$T_3=3(3+1)!-2\times 3!$

$\vdots$

$T_n=n(n+1)!-(n-1)n!$

Add, we get $\Sigma\ T_k=n(n+1)!$

6. Ans. (C)

Sol. Let required sum = S = coeff. of x^n in

$(1+x)^{3n}-{}^nC_1(1+x)^{3n-3}+{}^nC_2(1+x)^{3n-6}$

$-{}^nC_3(1+x)^{3n-9}+.....$

= Coeff. of x^n in $\{(1+x)^3-1\}^n$

= Coeff. of x^n in $x^n(x^2+3x+3)^n$

$=3^n$

7. Ans. (A)

Sol. Let $S=\Sigma\Sigma(i+j)(C_i^2+C_j^2+C_iC_j)$

Replacing i by n–i and j by n-j

$S=\Sigma\Sigma(n-i+n-j)(C_{n-i}^2+C_{n-j}^2+C_{n-i}.C_{n-j})$

$\Rightarrow\ 2S=2n\Sigma\Sigma(C_i^2+C_j^2+C_i.C_j)$

For More Material Join: @JEEAdvanced_2024

$$S = n\left\{n.{}^{2n}C_n + \frac{4^n - {}^{2n}C_n}{2}\right\} = n\frac{\{(2n-1){}^{2n}C_n + 4^n\}}{2}$$

8. Ans. (B,C)

Sol. $\sum_{r=0}^{n}\frac{r}{{}^nC_r} = \sum_{r=0}^{n}\frac{n-r}{{}^nC_{n-r}} = \frac{n}{2}\sum_{r=0}^{n}\frac{1}{{}^nC_r}$

$\Rightarrow \frac{n}{2} = \frac{n^2-4n+6}{2} \Rightarrow n = n^2 - 4n + 6$

$\Rightarrow n^2 - 5n + 6 = 0 \quad \Rightarrow n = 2, 3$

9. Ans. (A,C)

Sol. $\frac{m-1}{2}$ = integer $\Rightarrow m = 2$(integer) + 1

$\therefore$ m must be odd

Number of ways of formation of subsets B and C is

$= {}^mC_1 \cdot {}^{m-1}C_{\frac{m-1}{2}} = m.{}^{(m-1)}C_{\frac{m-1}{2}}$

(${}^mC_1 \rightarrow$ Number of selection of common element,

${}^{m-1}C_{\frac{m-1}{2}} \rightarrow$ Distribution of $(m-1)$ elements into two equal parts to B & C

10. Ans. (A,B,C,D)

Sol. Number of terms in the expansion of $(x_1 + x_2 + ...x_r)^n$ is ${}^{n+r-1}C_{r-1}$

(A) Number of terms in $(2x - 3y + z)^{10}$ is ${}^{10+3-1}C_{3-1}$=66

(B) Put $x = \frac{1}{2}, y = -\frac{1}{3}$ & z=1 we get

$$\sum_{\substack{0\le\alpha,\beta,\gamma\le10\\ \alpha+\beta+\gamma=10}} \frac{10!}{\alpha!\beta!\gamma!} = 3^{10}$$

(C) Cofficient of $x^2y^3z^5 = \frac{10!}{2!3!5!}\,2^2.(-3)^3.(1)^5$

$= -\frac{10!(9)}{5!}$

(D) Put x=1, y=1 & z=1

$$\sum_{\substack{0\le\alpha,\beta,\gamma\le10\\ \alpha+\beta+\gamma=10}} \frac{10!}{\alpha!\beta!\gamma!}2^{\alpha}(-3)^{\beta} = 0$$

11. Ans. (A,B,D)

Sol. * at least one fruit is chosen.

$(4 + 1)(3 + 1)(2 + 1)(2 + 1) - 1 = 179$

* at least 2 fruits are chosen.

$(4 + 1)(3 + 1)(2 + 1)(2 + 1) - 1 - (4) = 175$

* at most 2 fruits can be chosen .

selection of No fruit $\Rightarrow$ 1 method

selection of 1 fruit $\Rightarrow$ 4 method.

selection of 2 fruits $\Rightarrow 4 + {}^4C_2 = 10$

$= 1 + 4 + 10 = 15$

Paragraph for Question 12 to 14

12. Ans. (D)

Sol. A(1, 2) indicates k = 2

The man has to climb 6 steps

(i) 1, 1, 1, 1, 1, 1 $= {}^6C_0$ ways = 1

(ii) 1, 1, 1, 1, 2 $= {}^5C_1$ ways = 5

(iii) 1, 1, 2, 2 $= {}^4C_2$ ways = 6

(iv) 2, 2, 2 $= {}^3C_3$ ways = 1

$\therefore$ Total ways = 13

13. Ans. (C)

Sol. A(1, k) denotes the number of ways in which the man has to climb either 1 step or k steps at a time

He has to climb 3k steps

(i) $\underbrace{1,1,1,.....,1}_{3k\text{ times}}$ = 1 way

(ii) $\underbrace{k,1,1,.....,1}_{2k\text{ times}} = {}^{2k+1}C_1$ ways

(iii) $\underbrace{k,k,1,1,1,.....,1}_{k\text{ times}} = {}^{k+2}C_2$ ways

(iv) k, k, k = 1 way

$\therefore$ Total number of ways

$\Rightarrow 1 + {}^{2k+1}C_1 + {}^{k+2}C_2 + 1 = 26$

$\therefore$ k = 4

14. Ans. (A)

Sol. $\because \lambda_k = A(1, k+1) - A(1, k)$

$= 1 + {}^{2k+3}C_1 + {}^{k+3}C_2 + 1 -$

$\left(1 + {}^{2k+1}C_1 + {}^{k+2}C_2 + 1\right)$

$= 2k + 3 + \frac{(k+3)(k+2)}{2} - (2k + 1) -$

$\frac{(k+2)(k+1)}{2}$

$= 2 + \frac{k+2}{2}(k + 3 - k - 1)$

$\lambda_k = k + 4$

For More Material Join: @JEEAdvanced_2024

$\lambda_{k+1} - \lambda_k = k + 5 - (k + 4) = 1$

SECTION–II

1. Ans. 5.00

Sol. Let $\sum_{r=0}^{10} {}^{10}C_r \cdot (r-5)^3 = S$

replace r by 10 – r

$\Rightarrow \sum_{r=10}^{0} {}^{10}C_{10-r} \cdot (10-r-5)^3 = S$

$\Rightarrow \sum_{r=0}^{10} {}^{10}C_r (5-r)^3 = S$

$\Rightarrow 2S = 0 \Rightarrow S = 0$

$\Rightarrow \sum_{r=0}^{10} {}^{10}C_r (r-5)^3 = 0$

$\Rightarrow {}^{10}C_0(-5)^3 + \sum_{r=1}^{10} {}^{10}C_r (r-5)^3 = 0$

$\Rightarrow \sum_{r=1}^{10} {}^{10}C_r (r-5)^3 = 125$

2. Ans. 0.00

Sol. $(57)^{n+3} = (5 \times 11 + 2)^{n+3} = a$ multiple of $11 + 2^{n+3}$

$(79)^{n+1} = (7 \times 11 + 2)^{n+1} = a$ multiple of $11 + 2^{n+1}$

$(46)^n = (4 \times 11 + 2)^n = a$ multiple of $11 + 2^n$

$(57)^{n+3} + (79)^{n+1} + (46)^n = 11k + 2^n(2^3 + 2^1 + 1)$

for some $k \in N$

$= 11k + 11.2^n = 11(k + 2^n)$

Therefore $(57)^{n+3} + (79)^{n+1} + (46)^n$ is divisible by 11.

3. Ans. 196.00

Sol. Number of ways = No condition – (Two friends always together)

$= {}^{10}C_5 - {}^8C_3 = 252 - 56 = 196$

4. Ans. 8.00

Sol. $\because$ $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$ and

$\because$ the divisors are odd and not a multiple of 3

$\therefore$ the number of such divisors $= (2+1)\cdot(1+1) = 6$

and they are 1, 5, 7, 25, 35, 175

out of them 5, 35 are of type 3n – 1 and 1, 7, 25, 175 are of type 3n + 1

$\Rightarrow S_1 = 208, S_2 = 40$

$\Rightarrow$ remainder obtained by dividing S_1 by S_2 is 8.

Rankers Academy JEE

For More Material Join: @JEEAdvanced_2024